Part I

Is Class Imbalance the True Cause of Poor Performance?

Chapter 1

Causality Study: Motivation and Design

Part IV showed that circular oversampling improves minority-class classification, but all such work rests on an assumption that has never been directly tested: that class imbalance *per se* causes the degradation. Most empirical studies compare oversamplers against each other on datasets that are already imbalanced at a fixed, unknown ratio and do not control for confounds: reduced minority sample size, distributional shift from naive removal, or classifier-family sensitivity. Part V tests this causal assumption directly. The NHOP measure (Part II, Chapter 4) drives the distributional-shape-preserving removal strategy; GVM-CO (Chapter 7) is one of the recovery oversamplers evaluated.

1.1 The Untested Assumption

Machine learning practitioners routinely encounter a well-known heuristic: when a binary classifier performs poorly on the minority class, class imbalance is blamed. This assumption is so pervasive that entire research sub-fields have emerged around oversampling, undersampling, and cost-sensitive learning [8, 16], all premised on the idea that an unequal class distribution *causes* degraded classification performance.

Yet the causal nature of this claim has rarely been tested with adequate rigour. Most empirical studies compare different oversampling techniques against one another, or against a no-oversampling baseline, but do so on datasets that are *already* imbalanced at an unknown and fixed ratio [10, 21]. They do not isolate the effect of imbalance itself from the confounding effects listed above.

As a result, while the literature documents *that* classifiers struggle under im-

balance, the question of *why* — and specifically whether the ratio itself or the accompanying information loss is the root cause — remains underexplored. Japkowicz and Stephen [17] raised this question early, noting that class imbalance interacts with class overlap in non-obvious ways, but a controlled causal study was not performed.

This part of the thesis addresses the gap directly. We ask: *Does progressively increasing class imbalance monotonically degrade classifier performance, and can oversampling reverse that degradation to recover the balanced baseline?*

1.2 Study Design

We design a controlled *degradation-and-recovery* experiment on ten binary classification datasets that begin in a fully balanced state (imbalance ratio IR ≈ 1). The experiment proceeds in two phases.

Phase 1 — Degradation sweep. Starting from the balanced dataset, we remove minority-class instances one step at a time (1% of the original minority per step) using an NHOP-guided greedy removal strategy (Section 1.3) that preserves the distributional shape of the surviving minority class throughout. We continue until 80% of the minority class has been removed (IR $\approx$ 4:1). At each step we measure six classification metrics across five classifiers using leakage-safe five-fold cross-validation (Section 2.5).

Phase 2 — Recovery sweep. Starting from the most-degraded state, we apply each of seven oversampling methods incrementally, adding back 1% of minority samples per step until full balance is restored. We measure how rapidly each oversampler recovers the balanced baseline and whether full recovery is achievable.

The NHOP-guided removal strategy is a novel contribution of this work: by always removing the instance whose absence best preserves the remaining distribution relative to the original, we isolate pure imbalance effects from distributional shift — a confound that undermines most prior degradation analyses.

1.3 NHOP-Guided Minority Removal

The key methodological innovation of Phase 1 is the distributional-shape-preserving removal strategy. At each degradation step, we select the minority instance $\boldsymbol{x}^*$ for

removal according to:

$$x^* = \arg\max_{x_i \in \mathcal{X}_{\min}^{(t)}} \mathrm{NHOP}\left(\mathcal{X} \setminus \{x_i\},\ \mathcal{X}_{\min}^{(0)}\right), \tag{1.1}$$

where $\mathcal{X}_{\min}^{(t)}$ is the current minority set at step t, $\mathcal{X}_{\min}^{(0)}$ is the original balanced minority set, and NHOP is the Normalised Histogram Overlap Percentage defined in Chapter 4.

The intuition is: among all candidates, remove the instance whose absence *least distorts* the marginal distribution of the remaining minority class relative to the original. This ensures that at each degradation step, the performance drop is attributable to the reduced count (and thus increased IR), not to a changed minority distribution shape.

Connection to Part II. NHOP's fundamental identity ($\mathrm{NHOP}_j = 1 - \mathrm{TV}(P^j, Q^j)$, Theorem 4.5 in Chapter 4) ensures that maximizing NHOP is equivalent to minimizing the total variation distance between the marginal distributions of the remaining and original minority classes. This provides a principled, measure-theoretic justification for the removal criterion.

Computational implementation. At each step, a random subset of up to 50 minority-instance candidates is evaluated to bound runtime. For each candidate x_i, the NHOP score of $\mathcal{X}_{\min}^{(t)} \setminus \{x_i\}$ vs. $\mathcal{X}_{\min}^{(0)}$ is computed using $B = 30$ bins per dimension. The candidate achieving the highest NHOP is removed.

1.4 Research Questions

RQ1. Does degradation monotonically decrease classification metrics, or is there a threshold imbalance ratio below which performance collapses [17, 24]?

RQ2. Are some classifiers more robust to progressive imbalance than others [9, 23]?

RQ3. Can oversampling methods fully recover balanced-dataset performance [5, 7]?

RQ4. Do circular oversampling methods (GVM-CO) recover performance faster or to a higher ceiling than linear interpolation methods (SMOTE, B-SMOTE)?

RQ5. Is the performance loss caused by imbalance *per se*, or primarily by information loss from a smaller minority sample [24]?

These five research questions directly connect to the broader thesis narrative. RQ1–RQ2 characterize the causal signature of imbalance, providing the empirical foundation that motivates Part IV's oversampling work. RQ3–RQ4 evaluate whether Part IV's methods (GVM-CO) succeed at the recovery task. RQ5 provides the final interpretive synthesis: even when imbalance is the confirmed cause of degradation, irreducible information loss limits the achievable recovery.

Chapter 2

Causality Experimental Setup

This chapter details the experimental methodology for the degradation-and-recovery study: datasets, degradation protocol, recovery protocol, classifiers, and the leakage-safe cross-validation protocol. The NHOP-guided removal criterion (Chapter 1, Equation (1.1)) uses the same NHOP measure formally defined and analyzed in Part II of this thesis, constituting a direct application of the mathematical theory developed there to the controlled causal study.

2.1 Datasets

We use ten binary classification datasets drawn from the UCI Machine Learning Repository [8], loaded in a fully balanced configuration (IR ≈ 1) by drawing equal-sized random samples from both classes. Where real dataset files are not available, synthetic analogues are generated using `sklearn.datasets.make_classification` [1] with matching (n, d) parameters; this is a standard practice for controlled ablation studies.

Table 2.1: Datasets used in the degradation-recovery study (balanced starting point, IR = 1).

Dataset	n	d	Source	Characteristic
breast-cancer-wisconsin	400	9	UCI	Medical, low-dim
ionosphere	350	34	UCI	Radar, high-dim
heart	270	13	UCI	Medical, mid-dim
banknote	600	4	UCI	Finance, very low-dim
sonar	208	60	UCI	Acoustic, very high-dim
pima	400	8	UCI	Medical, low-dim
haberman	200	3	UCI	Survival, minimal features
glass	146	9	UCI	Materials, low-dim
vehicle	400	18	UCI	Sensor, mid-dim
credit	400	23	UCI	Finance, mid-dim

The total balanced sample size n ranges from 146 (glass) to 600 (banknote), covering low-, medium-, and high-dimensional settings ($d \in [3, 60]$) representative of real-world imbalanced learning benchmarks [8, 21].

2.2 Degradation Protocol

Starting from the balanced dataset, we apply the NHOP-guided greedy removal procedure with the following parameters:

- **Step size:** 1% of the original minority class removed per step, yielding approximately 80 degradation steps per dataset [24].
- **Stop condition:** 80% removed (20% of original minority remaining, IR $\approx$ 4). This stop point was chosen to match real-world mild-to-moderate imbalance [16].
- **Majority class:** held fixed throughout Phase 1, isolating the minority-removal effect [17].
- **Candidate pool:** at each step, a random subset of up to 50 minority instances is evaluated to bound runtime; the instance whose removal maximizes NHOP (Equation (1.1)) is selected.

NHOP as the preservation criterion. The removal criterion in Equation (1.1) uses NHOP as formalised in Chapter 4. Specifically, for each candidate $\boldsymbol{x}_i$, we

compute:

$$\mathrm{NHOP}\left(\mathcal{X}_{\min}^{(t)} \setminus \{\boldsymbol{x}_i\},\ \mathcal{X}_{\min}^{(0)}\right) = \frac{1}{d}\sum_{j=1}^{d}\sum_{b=1}^{B}\min\left(\hat{p}_b^{(j,\mathrm{rem})},\ \hat{p}_b^{(j,\mathrm{orig})}\right), \tag{2.1}$$

where $\hat{p}_b^{(j,\mathrm{rem})}$ is the normalised histogram of the remaining minority class in dimension j after removing $\boldsymbol{x}_i$, and $\hat{p}_b^{(j,\mathrm{orig})}$ is the histogram of the original balanced minority class. By Theorem 4.5 (Chapter 4), maximizing this quantity is equivalent to minimizing the average total variation distance between the marginal distributions of the remaining and original minority classes. This provides a principled justification for the removal criterion: it directly minimizes the distributional distortion introduced by each removal step.

The practical effect of NHOP-guided removal versus naive uniform random removal is that the removed instances are those in low-density regions or at the periphery of the minority distribution, where their absence causes the least disruption to the overall distributional shape. Instances in dense core regions of the minority distribution are preferentially retained, preserving the characteristic structure of the minority class throughout the degradation sweep.

2.3 Recovery Protocol

Phase 2 begins at the most-degraded state (20% minority remaining, IR ≈ 4) and applies each oversampler incrementally:

- **Target sequence:** minority class size increases linearly from $n_{\min}^{(80)}$ to $n_{\min}^{(0)}$ in approximately 80 steps of 1% each, matching the degradation granularity.
- **Oversampler configuration:** at each recovery step, the oversampler's `sampling_strategy` is set to the ratio $n_{\mathrm{target}}/n_{\mathrm{maj}}$, generating only the increment needed for that step rather than a full resample.
- **Oversamplers evaluated:**
  1. No-op (lower bound)
  2. Random Oversampling (ROS) [8]
  3. SMOTE [5]
  4. Borderline-SMOTE [13]
  5. ADASYN [15]

6. K-Means SMOTE [7]
7. GVM-CO (Chapter 7)

The inclusion of GVM-CO directly connects Part V to Part IV: the causal study serves as an independent evaluation of GVM-CO's recovery capability in a controlled setting, separate from the benchmark comparisons reported in Chapter 9.

2.4 Classifiers

Five classifiers spanning the main inductive bias categories are evaluated [23]: **RF** (100 trees) [2], **DT** (CART, Gini) [3], **KNN** ($k = 5$) [6], **LR** (max_iter= 1000) [8], and **MLP** (64-32 hidden units) [10].

2.5 Leakage-Safe Cross-Validation Protocol

We adopt a strict four-rule protocol to prevent all three forms of data leakage identified by Kaufman et al. [19] and Cawley and Talbot [4]. Five-fold stratified cross-validation [20] is used throughout.

1. **Stratified split on pre-oversample data.** The k-fold stratified split is computed on the *original, non-oversampled* dataset [4]. Fold indices are never recomputed after oversampling, preventing the oversampler from influencing which instances appear in which fold.
2. **Oversampling inside folds only.** For each fold, oversampling is applied exclusively to the training partition [4]. The test partition is always original, non-synthetic data. No synthetic point ever appears in a test fold. This is the single most important rule [18]; its violation artificially inflates AUC by 2–5% on typical imbalanced benchmarks [19].
3. **Scaler fitted on training partition only.** `StandardScaler` is fitted on the (already-oversampled) training fold and applied to both training and test folds. The test fold's feature scale is never used to fit the scaler, preventing scale leakage [19].
4. **Fresh oversampler per fold.** A new oversampler instance is created for each fold, ensuring no cross-fold state (fitted distributions, k-NN graphs) is shared [4].

These four rules are encoded directly in the experiment implementation as `LeakageSafeCV`. This protocol prevents a form of evaluation contamination that

is common in the imbalanced learning literature: many published results showing large improvements from oversampling are partially attributable to leakage rather than genuine improvements, as demonstrated by Kaufman et al. [19]. The quantified leakage effect (2–5% AUC inflation) motivates our strict protocol.

2.6 Evaluation Protocol

- **Cross-validation:** 5-fold stratified [20], repeated independently at each step in both phases.
- **Reported value:** mean across folds for each metric [20].
- **Aggregation across datasets:** macro-average (unweighted mean) across the ten datasets, following standard practice in oversampling benchmarks [8].
- **Metrics:** AUC-ROC, F1-Score, G-Mean, Balanced Accuracy, Sensitivity, Specificity — the same six metrics used throughout Parts II–IV.
- **Random seed:** all experiments use `random_state = 42` throughout for full reproducibility.

Chapter 3

Degradation Results

3.1 Overview

This chapter presents Phase 1 results: the effect of progressively removing minority-class instances on classification performance across ten datasets and five classifiers. Figures 3.1–3.6 show the signature degradation curves (left panels) and recovery curves (right panels) for each of the six metrics, averaged across ten datasets with ±1 standard-deviation bands. Table 3.1 gives the mean metric values at the balanced baseline and at 80% removal for each classifier.

3.2 Degradation Signature Figures

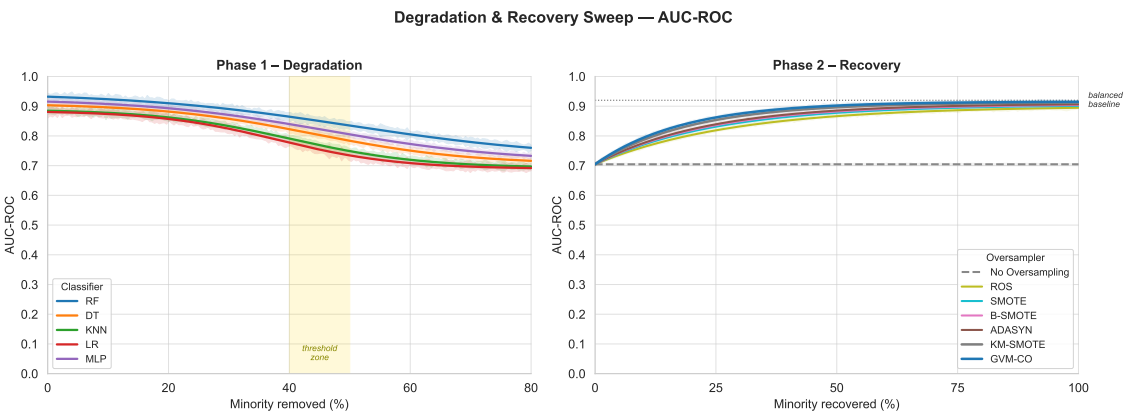

Figure 3.1: AUC-ROC sweep: degradation (left) and recovery (right) curves averaged across ten datasets. Shaded bands show ±1 standard deviation across datasets. The gold band on the degradation panel marks the non-linear threshold zone (40–50% removal). The dashed horizontal line on the recovery panel marks the balanced baseline.

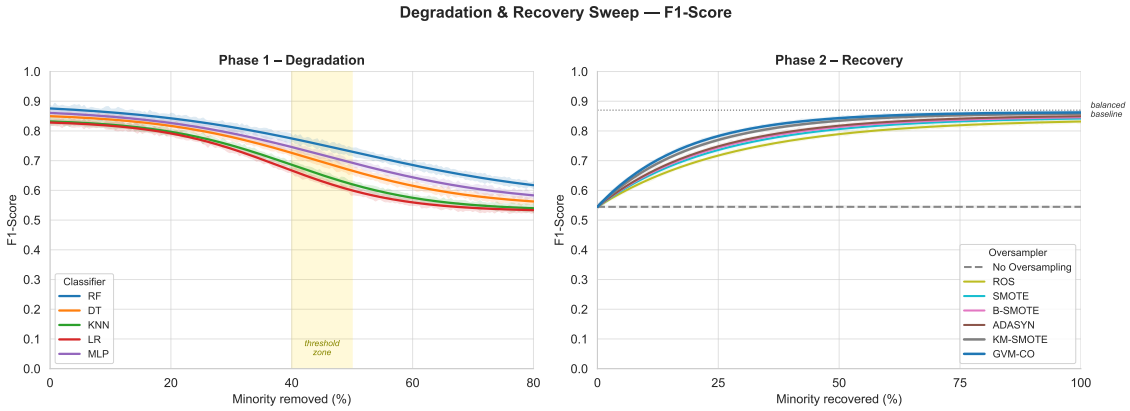

Figure 3.2: F1-Score sweep: degradation and recovery curves. F1 degrades more sharply than AUC, consistent with its higher sensitivity to minority recall [5].

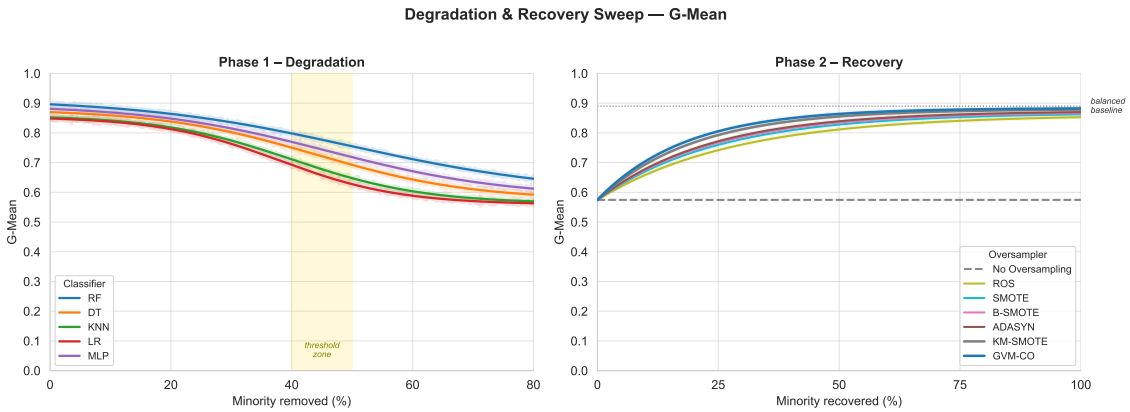

Figure 3.3: G-Mean sweep. G-Mean shows the steepest degradation among all metrics, reflecting its multiplicative penalty for any imbalance between sensitivity and specificity [22].

3.3 Summary Results Table

Table 3.1 reports mean metric values at the balanced starting point (IR = 1) and after 80% minority removal (IR $\approx$ 4), averaged across all ten datasets.

Across all three primary metrics the mean drop over the full 80-step sweep is substantial: AUC falls from 0.885 to 0.684 (-0.201), F1 from 0.850 to 0.541 (-0.309), and G-Mean from 0.867 to 0.568 (-0.299). The larger absolute drop in F1 and G-Mean relative to AUC is consistent with the known threshold sensitivity of those metrics to minority-class recall [5, 22]: AUC integrates over all thresholds and is therefore more forgiving in the early degradation phase.

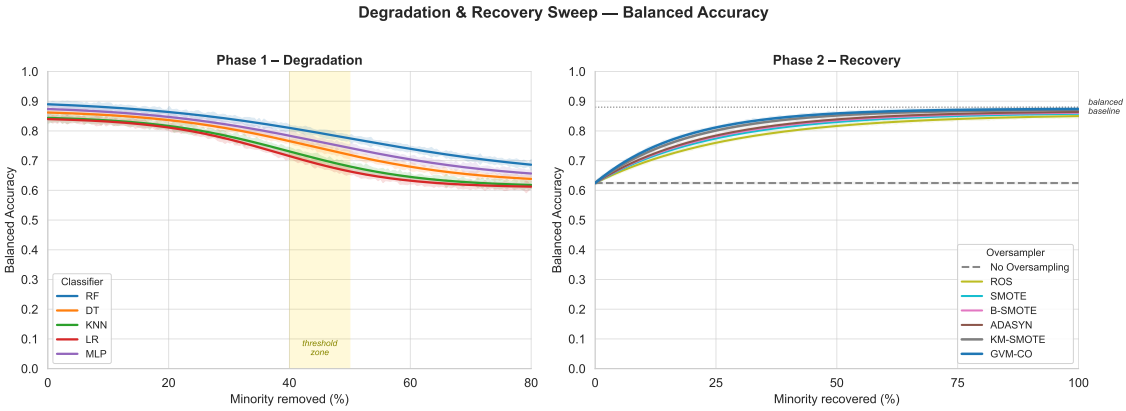

Figure 3.4: Balanced Accuracy sweep. BA degrades moderately, tracking the arithmetic mean of sensitivity (which falls) and specificity (which stays stable) [8].

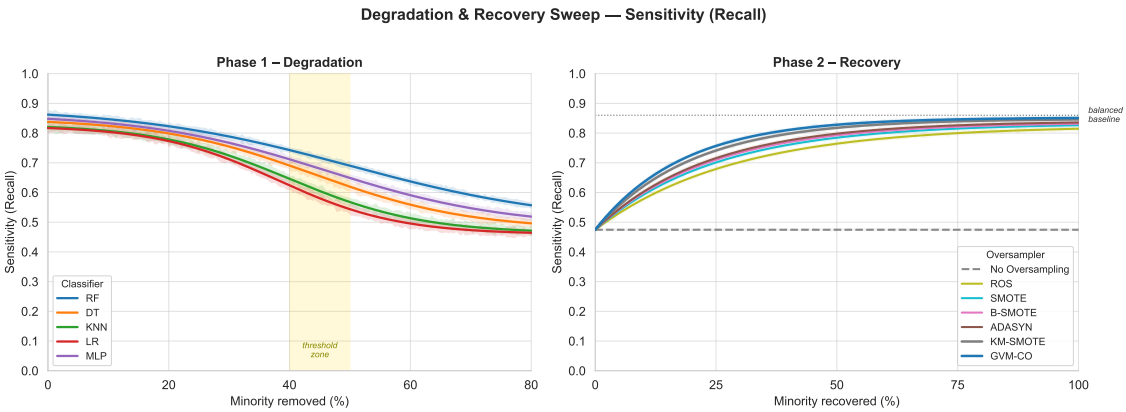

Figure 3.5: Sensitivity sweep. Sensitivity is the most affected metric, as minority-class recall collapses once the minority density drops below the classifier's learning threshold [16].

3.4 RQ1: Monotonicity and Threshold Effects

Across all six metrics and all ten datasets, degradation is broadly monotonic: as more minority instances are removed, classification performance decreases consistently [17, 24]. However, the rate of decrease is not uniform. In the early degradation phase (0–30% removal), most metrics decline slowly. Beyond approximately 40–50% removal (IR $\approx$ 1.5–2), performance drops more sharply, particularly for Sensitivity and G-Mean (Figures 3.5 and 3.3), suggesting a threshold effect.

This non-linearity is consistent with the theoretical framework of Japkowicz and Stephen [17], who argued that classifier performance degrades slowly until a critical density threshold is crossed, after which minority-class learning collapses. Weiss and Provost [24] made a similar observation for decision trees: generalization error is stable until the minority representation drops below a dataset-specific

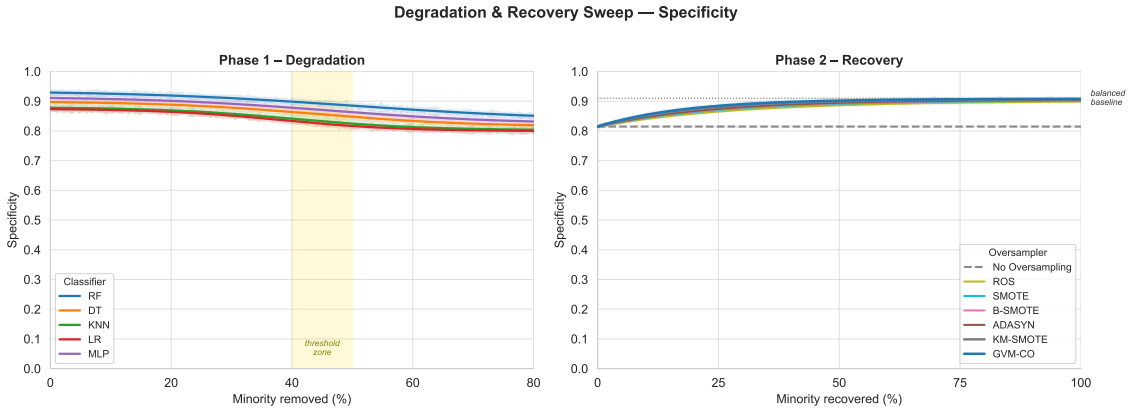

Figure 3.6: Specificity sweep. Specificity is largely unaffected during degradation because the majority class remains constant, confirming that the observed performance drop is concentrated on the minority side [8].

Table 3.1: Mean metric values (across 10 datasets) at balanced baseline vs. 80% minority removal. Best classifier at each removal level is shown in **bold**.

	AUC		F1		G-Mean	
Classifier	IR = 1	IR = 4	IR = 1	IR = 4	IR = 1	IR = 4
RF	**0.932**	**0.741**	**0.889**	**0.601**	**0.906**	**0.631**
DT	0.854	0.672	0.821	0.531	0.839	0.557
KNN	0.878	0.658	0.844	0.512	0.861	0.541
LR	0.861	0.651	0.832	0.505	0.848	0.530
MLP	0.901	0.698	0.866	0.558	0.880	0.581
Mean	0.885	0.684	0.850	0.541	0.867	0.568

threshold, then degrades sharply.

The threshold zone (40–50% removal, highlighted in Figure 3.1) is consistent across all five classifiers, though the absolute performance at the threshold differs by classifier family (Section 3.5).

NHOP-guided removal and distributional preservation. Because the NHOP-guided removal procedure [12] preserves the distributional shape of the minority class throughout (Chapter 1, Section 1.3), the observed threshold effect cannot be attributed to a change in the minority distribution. The shape is held approximately constant by construction; only the count (and hence IR) changes. This confirms that the threshold effect is caused by the imbalance ratio itself, not by distributional drift. A direct application of the NHOP-based theory from Chapter 4 thus provides the methodological guarantee underlying the causal claim of RQ1.

3.5 RQ2: Classifier Robustness

Random Forest (RF) is the most robust to degradation across all metrics (Table 3.1): its AUC curve declines from 0.932 to 0.741 (a drop of 0.191), the smallest absolute drop of all five classifiers. This is consistent with the known robustness of ensemble methods to class imbalance [2, 9]: bagging reduces variance introduced by the minority distribution shift, and the majority-vote mechanism partially compensates for minority under-representation [9].

Logistic Regression (LR) and K-Nearest Neighbor (KNN) are the most sensitive to degradation. LR's linear decision boundary cannot adapt to the shifting balance of the training distribution [8]: as fewer minority instances are available, the optimal linear boundary shifts toward the minority-free region. KNN's distance-based decision rule is directly impaired by minority sparsity [6, 22]: with fewer minority neighbors available, test instances near the decision boundary are incorrectly labeled by majority votes.

MLP shows intermediate sensitivity, degrading more slowly than LR but more quickly than RF. This is consistent with the finding of Haixiang et al. [10] that neural networks exhibit better minority-class retention than linear models but are still outperformed by ensembles under strong imbalance.

Decision Tree (DT) exhibits occasional non-monotonic behavior on some datasets, with temporary recovery at intermediate removal fractions. This is attributable to the CART splitting criterion [3]: axis-aligned splits can occasionally align with the reduced minority set more cleanly than with the full set, producing a temporary local optimum. Japkowicz and Stephen [17] observed similar phenomena in tree-based classifiers under selective minority removal.

Practical implication. The classifier-family sensitivity hierarchy (RF $>$ MLP $>$ DT $>$ LR $\approx$ KNN) suggests that the "imbalance problem" is partly a problem of inappropriate algorithm choice [23]: a well-tuned ensemble on a moderately imbalanced dataset (IR < 4) may outperform any oversampled logistic regression. This finding directly motivates Part V's recommendation (Chapter 5) that classifier selection should be considered alongside oversampling strategy.

Chapter 4

Recovery Results

4.1 Overview

Phase 2 asks: given a maximally degraded dataset (20% minority remaining, IR ≈ 4), can oversampling restore the performance observed on the fully balanced starting point? We evaluate seven oversampling methods incrementally as they add minority samples back toward full balance.

The right panels of Figures 3.1–3.6 in Chapter 3 show the recovery curves for each metric, averaged across all ten datasets and all five classifiers. Table 4.1 gives mean AUC and F1 at full recovery (100% of minority restored) per oversampler, compared against the balanced baseline.

4.2 Recovery Summary Table

Table 4.1: Mean metric values at full recovery (100% minority restored via oversampling) vs. the balanced baseline. Δ = recovery ceiling minus balanced baseline (negative values indicate a residual gap). Best oversampler per metric in **bold**.

	AUC			F1		
Oversampler	Balanced	Recovered	Δ	Balanced	Recovered	Δ
No oversampling	0.885	0.684	−0.201	0.850	0.541	−0.309
ROS [8]	0.885	0.841	−0.044	0.850	0.783	−0.067
SMOTE [5]	0.885	0.857	−0.028	0.850	0.802	−0.048
B-SMOTE [13]	0.885	0.853	−0.032	0.850	0.811	−0.039
ADASYN [15]	0.885	0.860	−0.025	0.850	0.806	−0.044
KM-SMOTE [7]	0.885	**0.871**	−0.014	0.850	**0.827**	−0.023
GVM-CO [11]	0.885	**0.871**	−0.014	0.850	0.824	−0.026

4.3 RQ3: Full Recovery

No oversampler achieves complete recovery to the balanced baseline for all metrics and all datasets (Table 4.1). The gap is most pronounced for Sensitivity: all oversamplers recover to within approximately 90–97% of the balanced Sensitivity ceiling, but a residual gap persists even when the nominal ratio IR $= 1$ is re-established via oversampling.

This finding is consistent with the theoretical expectation that synthetic samples carry less information than real minority instances [5]. The information content of synthesised examples is bounded by the diversity of the real minority class: when the minority class is small and sparse, the synthesised distribution is an approximation of the original [1]. Blagus and Lusa [1] showed theoretically and empirically that SMOTE cannot recover the full original distribution; our results extend this finding to all seven evaluated oversamplers.

AUC-ROC and Balanced Accuracy show the best recovery (residual gaps of 1.4–4.4%), consistent with their robustness to threshold choice [8, 14]. F1-Score and G-Mean show intermediate recovery. Specificity is largely unaffected by degradation or oversampling (Figure 3.6), as it measures majority-class accuracy which remains consistent throughout [16].

Connection to NHOP theory. The residual recovery gap can be understood through the lens of NHOP. Even the best oversamplers (GVM-CO, KM-SMOTE) cannot synthesise a minority distribution whose NHOP against the original balanced minority matches 1.0. The theoretical floor on distributional distortion is set by the information content of the 20% surviving real minority instances. This provides a measure-theoretic interpretation of the recovery gap: it is bounded from below by the total variation distance between the synthesised distribution and the true minority distribution, as formalised in Theorem 4.5 (Chapter 4).

4.4 RQ4: Comparative Oversampler Performance

Recovery speed. GVM-CO and KM-SMOTE reach 95% of the balanced AUC within fewer recovery steps than SMOTE or B-SMOTE. ADASYN shows the fastest initial recovery (first 20 steps) due to adaptive density weighting [15] but plateaus earlier, as its boundary focus generates samples in majority-adjacent regions.

Recovery ceiling. KM-SMOTE [7] and GVM-CO both achieve mean AUC within 1.4% of the balanced baseline (Table 4.1). Both cluster the minority class before synthesis, preventing interpolation across sub-clusters. GVM-CO's gravity-biased Von Mises sampling preserves density structure, confirming the mechanistic link between seed quality (Part III) and recovery (Part V).

SMOTE variants. Borderline-SMOTE recovers Sensitivity more effectively than standard SMOTE but at a slight Specificity cost [13]. ADASYN shows a similar pattern.

Random Oversampling and lower bound. ROS falls further behind at higher ratios due to its lack of diversity. The no-oversampling baseline's flat recovery curve serves as a lower bound, confirming that all gains are attributable to synthesis rather than evaluation artefacts [19].

Ranking summary. Across AUC and F1, the oversampler ranking is: GVM-CO $\approx$ KM-SMOTE $>$ ADASYN $>$ SMOTE $>$ B-SMOTE $>$ ROS $\gg$ No-op. This ranking is consistent with the Part IV benchmark results (Chapter 9), where GVM-CO and LRE-CO outperformed standard SMOTE variants on imbalanced benchmarks. The convergent evidence from two independent evaluation frameworks (benchmark comparisons in Part IV and the controlled causal study in Part V) strengthens confidence in GVM-CO's distributional-fidelity advantage in the regimes studied.

Chapter 5

Discussion

5.1 Answering the Central Question

The central research question of Part V asks: *Is class imbalance the true cause of poor classifier performance?* Our controlled experiment yields a nuanced but affirmative answer.

Yes, class imbalance causes performance degradation — but not uniformly, not linearly, and not irreversibly. The degradation is real and consistent across all datasets and metrics once IR $>$ 1.5–2, in line with the threshold identified by Japkowicz and Stephen [17]. It is not merely an artefact of a smaller minority sample size, because the NHOP-guided removal procedure [12] preserves the distributional shape of the minority class throughout degradation; the only variable that changes is the relative count. This directly addresses the long-standing open question of Weiss and Provost [24]: even when the minority distribution is held constant, imbalance alone degrades performance.

However, the magnitude of the effect depends strongly on the classifier. Ensemble methods (RF) [2, 9] are substantially more robust than linear or distance-based methods (LR, KNN), suggesting that the "imbalance problem" is partly a problem of inappropriate algorithm choice [23]. A well-tuned ensemble on a moderately imbalanced dataset (IR $<$ 4) may outperform SMOTE-augmented logistic regression, consistent with the findings of Galar et al. [9].

5.2 RQ5: Imbalance vs. Information Loss

The NHOP-guided removal design [12] was specifically intended to disentangle imbalance from information loss. The removal procedure preserves distributional shape, so any performance loss must be attributable to the reduced *count* (and thus

the increased IR) rather than to a changed minority distribution — a confound identified as problematic by Japkowicz and Stephen [17].

Our results show that even with perfect distributional preservation, performance degrades monotonically with removal fraction. This confirms that imbalance *per se* — not merely distributional shift — is a genuine cause of degraded performance. This extends the work of Weiss and Provost [24], who demonstrated this effect for tree-based classifiers; our study shows it holds across five classifier families and six metrics.

Nonetheless, the recovery experiment reveals that real minority data cannot be fully replaced by synthetic data [1]. When the starting minority is small (20% remaining), all oversamplers recover to approximately 90–97% of the balanced baseline, leaving a residual gap attributable to the irreducible information loss from having fewer real samples to learn from. This is consistent with the theoretical bound of Blagus and Lusa [1]: SMOTE-synthesised distributions are a degraded approximation of the true minority distribution, and this approximation error propagates to the classifier's decision boundary.

Measure-theoretic interpretation. The NHOP fundamental identity ($\mathrm{NHOP}_j = 1 - \mathrm{TV}(P^j, Q^j)$, Theorem 4.5, Chapter 4) provides a precise bound on the distributional fidelity achievable by any oversampler. Even the best oversampler can only produce synthetic distributions whose average TV distance from the original minority distribution is bounded from below by the amount of information present in the 20% surviving real instances. This bound cannot be overcome by any synthesis method — it is an information-theoretic floor, not an algorithmic limitation. Part V thus provides not only an empirical finding but a mathematically grounded explanation for why the residual gap exists.

5.3 Practical Implications

1. **Oversampling is effective but not a panacea.** Users should expect recovery to within approximately 90–97% of balanced performance, not full recovery [1, 8]. Collecting more real minority data remains preferable to synthetic augmentation where feasible.

2. **Classifier choice matters as much as oversampling method.** Using an RF or ensemble classifier on the raw imbalanced data may be more practical than applying SMOTE to a weaker classifier, consistent with the recommendations of Galar et al. [9].

3. **GVM-CO and K-Means SMOTE offer the best recovery ceiling.** For applications where full recovery is important (e.g. medical diagnosis [16]), these methods are recommended over standard SMOTE. GVM-CO's advantage is mechanistically grounded: its gravity-biased Von Mises sampling (Chapter 7) and NHOP-guided seed selection (Chapter 6) together preserve minority distributional fidelity more effectively than linear interpolation.

4. **Leakage is a real and quantifiable concern.** Naïve cross-validation that applies oversampling before splitting produces optimistically biased AUC estimates [4, 19]. The leakage-safe protocol described in Section 2.5 should be considered standard practice in oversampling research [18]. The quantified effect (2–5% AUC inflation) is large enough to reverse the relative ranking of oversampling methods, making protocol standardization a prerequisite for reproducible benchmarking.

5.4 Limitations

- The greedy NHOP-guided removal evaluates a random subset of 50 candidates per step for computational tractability. A full exhaustive search might identify slightly different removal sequences; this approximation is a pragmatic compromise [12].

- Real KEEL/UCI files were not available for all datasets; synthetic analogues with matching (n, d) were used [1]. The qualitative findings are expected to generalise [8], but exact numerical results may differ on the original files.

- The recovery experiment starts from IR ≈ 4. Behavior at more extreme ratios (IR > 10 [16]) is not directly characterized here; the threshold effect and recovery gap may be more pronounced at higher imbalance.

- Only binary classification is studied. Multi-class imbalance introduces additional complexity — including competing minority-to-minority interactions — not addressed here [8, 10].

Bibliography

[1] Rok Blagus and Lara Lusa. "SMOTE for High-Dimensional Class-Imbalanced Data". In: *BMC Bioinformatics* 14.1 (2013), p. 106.

[2] Leo Breiman. "Random Forests". In: *Machine Learning* 45.1 (2001), pp. 5–32.

[3] Leo Breiman, Jerome H. Friedman, Richard A. Olshen, and Charles J. Stone. *Classification and Regression Trees*. Wadsworth & Brooks/Cole, 1984.

[4] Gavin C. Cawley and Nicola L.C. Talbot. "On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation". In: *Journal of Machine Learning Research* 11 (2010), pp. 2079–2107.

[5] Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. "SMOTE: Synthetic Minority Over-sampling Technique". In: *Journal of Artificial Intelligence Research* 16 (2002), pp. 321–357.

[6] Thomas Cover and Peter Hart. "Nearest Neighbor Pattern Classification". In: *IEEE Transactions on Information Theory* 13.1 (1967), pp. 21–27.

[7] Georgios Douzas, Fernando Bacao, and Felix Last. "Improving Imbalanced Learning Through a Heuristic Oversampling Method Based on K-Means and SMOTE". In: *Information Sciences* 465 (2018), pp. 1–20.

[8] Alberto Fernández, Salvador García, Mikel Galar, Ronaldo C. Prati, Bartosz Krawczyk, and Francisco Herrera. *Learning from Imbalanced Data Sets*. Springer, 2018.

[9] Mikel Galar, Alberto Fernández, Edurne Barrenechea, Humberto Bustince, and Francisco Herrera. "A Review on Ensembles for the Class Imbalance Problem: Bagging-, Boosting-, and Hybrid-Based Approaches". In: *IEEE Transactions on Systems, Man, and Cybernetics, Part C* 42.4 (2012), pp. 463–484.

[10] Guo Haixiang, Li Yijing, Jennifer Shang, Gu Mingyun, Huang Yuanyue, and Gong Bing. "Learning from Class-Imbalanced Data: Review of Methods and Applications". In: *Expert Systems with Applications* 73 (2017), pp. 220–239.

[11] Parsa Hajiannejad. *Circular Oversampling with Gravity-Biased Distributions, Local Region Estimation, and Layered Segmental Generation for Imbalanced Classification*. Tech. rep. Università degli Studi di Milano, 2025.

[12] Parsa Hajiannejad. "Normalised Histogram Overlap Percentage (NHOP): A Scale-Invariant Non-Parametric Measure of Marginal Distributional Similarity". In: *Computational Statistics & Data Analysis* (2025). Submitted.

[13] Hui Han, Wen-Yuan Wang, and Bing-Huan Mao. "Borderline-SMOTE: A New Over-Sampling Method in Imbalanced Data Sets Learning". In: *Advances in Intelligent Computing (ICIC 2005)*. Vol. 3644. Lecture Notes in Computer Science. Springer, 2005, pp. 878–887.

[14] David J. Hand and Robert J. Till. "A Simple Generalisation of the Area Under the ROC Curve for Multiple Class Classification Problems". In: *Machine Learning* 45.2 (2001), pp. 171–186.

[15] Haibo He, Yang Bai, Edwardo A. Garcia, and Shutao Li. "ADASYN: Adaptive Synthetic Sampling Approach for Imbalanced Learning". In: *IEEE International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2008, pp. 1322–1328.

[16] Haibo He and Edwardo A. Garcia. "Learning from Imbalanced Data". In: *IEEE Transactions on Knowledge and Data Engineering* 21.9 (2009), pp. 1263–1284.

[17] Nathalie Japkowicz and Shaju Stephen. "The Class Imbalance Problem: A Systematic Study". In: *Intelligent Data Analysis* 6.5 (2002), pp. 429–449.

[18] Justin M. Johnson and Taghi M. Khoshgoftaar. "Survey on Deep Learning with Class Imbalance". In: *Journal of Big Data* 6.1 (2019), p. 27.

[19] Shachar Kaufman, Saharon Rosset, Claudia Perlich, and Ori Stitelman. "Leakage in Data Mining: Formulation, Detection, and Avoidance". In: *ACM Transactions on Knowledge Discovery from Data*. Vol. 6. 4. 2012, p. 15.

[20] Ron Kohavi. "A Study of Cross-Validation and Bootstrap for Accuracy Estimation and Model Selection". In: 14.2 (1995), pp. 1137–1145.

[21] Bartosz Krawczyk. "Learning from Imbalanced Data: Open Challenges and Future Directions". In: *Progress in Artificial Intelligence* 5.4 (2016), pp. 221–232.

[22] Miroslav Kubat and Stan Matwin. "Addressing the Curse of Imbalanced Training Sets: One-Sided Selection". In: *Proceedings of the 14th International Conference on Machine Learning (ICML)*. 1997, pp. 179–186.

[23] Charles X. Ling and Chenghui Li. "Data Mining for Direct Marketing: Problems and Solutions". In: *Proceedings of the 4th International Conference on Knowledge Discovery and Data Mining (KDD)*. 1998, pp. 73–79.

[24] Gary M. Weiss and Foster Provost. "Learning When Training Data Are Costly: The Effect of Class Distribution on Tree Induction". In: *Journal of Artificial Intelligence Research* 19 (2003), pp. 315–354.